

KUSHI HYGIEIA HEALTHCARE PLATFORM

Technical Architecture, Research Benchmarks, Clinical Datasets, and System References

REPOSITORY

EdlaAbhishek/Kushi-hyeieia

LIVE PRODUCTION

kushi-hygieia.vercel.app

DOCUMENT VERSION

v1.0.0 (Release Edition)

CLASSIFICATION

Open Architecture & Research

1. Platform Overview & Etymology

Kushi Hygieia is an enterprise-grade healthcare platform that unifies hybrid quantum machine learning (QML), 24/7 conversational clinical artificial intelligence, digital OPD queue management, and real-time hospital bed intelligence into a cohesive medical ecosystem.

Core Etymology & Vision

Khushi / Kushi: Sanskrit and Indian root representing happiness, vitality, and well-being.

Hygieia: The ancient Greek goddess of good health, sanitation, cleanliness, and the prevention of disease.

Vision: Transforming complex health data into personalized AI-powered insights, connecting individuals with the right medical support, and making healthcare more accessible, informed, and connected.

2. Biomedical Datasets & Clinical Benchmarks

The platform's **Quantum Disease Intelligence** screening system evaluates disease risk across four verified, open-access medical databases. Preprocessing transformers and PCA reductions are strictly fitted on training splits to prevent data leakage.

DATASET NAME	CLINICAL DOMAIN	SOURCE / CONTRIBUTORS	SAMPLES & FEATURES	TASK & TARGET	REFERENCE REPOSITORY
Breast Cancer Wisconsin	Oncology (Biopsy FNA)	Univ. of Wisconsin Hospitals (Wolberg, Street, Mangasarian)	569 samples, 30 continuous features	Binary (Benign: 0, Malignant: 1)	mswamyvvce/Quantum-Breast-Cancer-Detection
Pima Indians Diabetes	Metabolic Endocrinology	NIDDK / UCI Machine Learning Repository	768 samples, 8 medical markers	Binary (Negative: 0, Diabetic: 1)	Anto4K/VQC_on_Pima_Indians_Diabetes_Dataset
Cleveland Heart Disease	Cardiology & CAD	Cleveland Clinic Foundation (Dr. Robert Detrano)	303 samples, 13 clinical attributes	Binary (Healthy: 0, Heart Disease: 1)	TirtheshJani/QML-Healthcare-Diagnostics
Oxford Parkinson's	Neurology (Acoustic Dysphonia)	Univ. of Oxford & NCVS (Dr. Max Little et al.)	195 recordings, 22 acoustic metrics	Binary (Control: 0, Parkinson's: 1)	ranazsaad/QMedicine-Parkinson-Detection

Breast Cancer Wisconsin (Diagnostic)

Oncology Benchmark

Measurements computed from digitized Fine Needle Aspirate (FNA) images of breast masses describing characteristics of the cell nuclei present in the image (radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, fractal dimension). Evaluates high-dimensional quantum kernel boundaries against malignant formations.

UCI ML ID: 17 License: CC BY 4.0 Qubits: 4 Qubits Reduction: PCA Scaling: MinMaxScaler

Pima Indians Diabetes Database

Metabolic Health

Diagnostic metabolic predictors (oral glucose tolerance 2-hr, diastolic blood pressure, triceps skin fold, 2-hr serum insulin, BMI, diabetes pedigree score, and age). Zero-as-missing clinical imputation applied to biologically impossible zero values prior to variance-based feature selection.

UCI ML ID: 34 License: Public Domain Qubits: 4 Qubits Reduction: SelectKBest

Strategy: Zero-as-missing Imputation

3. Quantum Machine Learning (QML) Architecture

Kushi Hygieia implements a dual-mode quantum computational pipeline: an external Python-accelerated Qiskit execution engine and an internal, zero-dependency Node.js Quantum Simulation Engine for serverless cloud execution.

3.1 Quantum Software Stack

- IBM Qiskit SDK (`qiskit >= 1.0.0`):** Open-source quantum computing framework for circuit assembly, gate transpilation, and quantum state manipulation.
- Qiskit Machine Learning (`qiskit-machine-learning`):** Provides quantum kernel algorithms and variational quantum classifiers.
- Qiskit Aer Simulator (`qiskit-aer`):** High-performance quantum simulator supporting statevector evolution and shot-based quantum measurements (default: 1,024 shots).
- Scikit-Learn (`scikit-learn`):** Preprocessing pipelines (`StandardScaler` , `PCA` , `SelectKBest`) and classical baselines (`SVC` , `LogisticRegression` , `RandomForestClassifier`).

3.2 Circuit Mathematical Formulations

1. ZZFeatureMap (Nonlinear Pauli Expansion)

Encodes classical feature vectors $\mathbf{x}$ into high-dimensional quantum Hilbert state space using single-qubit rotations and two-qubit entangling gates:
$$U_{\Phi}(\mathbf{x}) = \exp\left(i \sum_j \Phi_{\{j\}}(\mathbf{x}_j) Z_j + i \sum_{\{j,k\}} \Phi_{\{j,k\}}(\mathbf{x}_j, \mathbf{x}_k) Z_j Z_k \right)$$
Utilizes 2 repetitions with linear CNOT entanglement to capture feature correlations inaccessible to classical hyperplanes.

2. Quantum Support Vector Classifier (QSVC) & Quantum Kernel

Calculates transition fidelity inner products between quantum states in the 2^n Hilbert space:
$$K(\mathbf{x}_i, \mathbf{x}_j) = \langle \theta^n | U_{\Phi}^{\dagger}(\mathbf{x}_j) U_{\Phi}(\mathbf{x}_i) | \theta^n \rangle^2$$
Constructs an $N \times N$ quantum kernel Gram matrix evaluated through quantum state measurement, then resolved using classical dual quadratic optimization.

3. Variational Quantum Classifier (VQC)

Combines parameterized quantum circuits (ansatz) with classical gradient-free optimizers (COBYLA / SPSA) to iteratively minimize cross-entropy loss over patient clinical embeddings.

4. Artificial Intelligence & Clinical LLM Orchestration

The platform integrates state-of-the-art Generative AI to provide patient guidance, triage support, and clinical document translation:

PROVIDER & LIBRARY	MODEL FAMILY	CLINICAL FEATURE IMPLEMENTATION
Google Gemini API (<code>@google/generative-ai</code>)	Gemini 2.5 / 1.5 Pro & Flash	Kushi Care AI: 24/7 conversational medical triage with empathy. Prescription OCR & Simplification: Translates medical jargon and dosage terms into clear daily schedules. Symptom Assessment: Evaluates reported pain, duration, and severity against triage urgency levels.
OpenRouter SDK (<code>@openrouter/sdk</code>)	Multi-Provider Fallback Orchestration	- Automated multi-model redundancy ensuring 99.99% availability. - Prevents conversational disruption during single-vendor upstream downtime.

Ethical Safety & Diagnostic Disclaimer Boundaries

All conversational AI outputs include mandatory medical safety disclaimers: preliminary triage recommendations are informational and educational, and are strictly explicitly declared as non-substitutes for professional clinical diagnosis. Immediate red-flag symptoms trigger instant emergency ambulance and hospital bed routing.

5. Frontend Engineering, UI Architecture & Telehealth

The user experience is engineered for rapid visual feedback, accessibility, and high aesthetic fidelity:

Component Layer	Technology / Library	Design Role & Implementation
Application Core	React 18 + Vite 6 + React Router v6	Modern Single Page Application (SPA), role-based auth routing for Patients, Doctors, and Administrators.
Styling & Design System	Tailwind CSS v4 + Custom HSL Tokens	Curated clinical palette: Medical Teal (#0D9488), Deep Blue (#0A2540), crisp glassmorphism cards.
Animations & Wave Engine	Framer Motion + Custom SVG Canvas	GPU-accelerated multi-layered undulating gradient wave canvas inspired by modern creative systems (21st.dev / Magic UI).
Iconography	Lucide React (lucide-react)	Accessible, consistent SVG medical, clinical, and administrative icons.
Data Visualization	Recharts (recharts)	Quantum basis state probability histograms, feature importance rankings, and historical patient vitals.
Telehealth Consultation	PeerJS (peerjs / WebRTC)	Peer-to-peer real-time encrypted video and audio consultation between patient and licensed doctor.
In-Browser Document OCR	PDF.js (pdfjs-dist) + Mammoth.js	Client-side parsing of medical laboratory test reports, scan summaries, and .docx prescriptions.
Client Search Engine	Fuse.js (fuse.js)	Instant zero-latency fuzzy search across hospital directories, bed availability, and medicines.

6. Cloud Infrastructure, Security & Healthcare Compliance

Kushi Hygieia adheres to modern health information architecture standards to safeguard patient data and ensure uninterrupted access:

- Database & Authentication:** Supabase (PostgreSQL 16) with Row-Level Security (RLS) policies enforcing zero data bleed between patient records, JSON Web Token (JWT) session validation, and encrypted storage buckets.
- DISHA (Digital Information Security in Healthcare Act - India):** Incorporates strict patient ownership of personal health records (PHR), granular data sharing permissions, and audit logging.
- ABDM (Ayushman Bharat Digital Mission):** System design aligns with India's national digital health architecture, including tokenized OPD appointment scheduling and digital health record federation.
- Data Encryption:** AES-256 encryption at rest for all stored reports and vitals; TLS 1.3 protocol encryption for all data in transit across client, server, and APIs.
- Serverless Compute:** Hosted on Vercel Global Edge Network with automatic CI/CD deployment and child-process sandboxing.

7. Academic Literature & Formal Citations (APA 7th Edition)

1. Wolberg, W. H., Street, W. N., & Mangasarian, O. L. (1995). Machine learning techniques to diagnose breast cancers from fine-needle aspirates. *Cancer Letters*, 77(2-3), 163-171. [https://doi.org/10.1016/0304-3835\(94\)90099-X](https://doi.org/10.1016/0304-3835(94)90099-X)

2. Smith, J. W., Everhart, J. E., Dickson, W. C., Knowler, W. C., & Johannes, R. S. (1988). Using the ADAP learning algorithm to forecast the onset of diabetes mellitus. In *Proceedings of the Annual Symposium on Computer Application in Medical Care* (p. 261). American Medical Informatics Association.

3. Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J., Sandhu, S., Guppy, K. H., Lee, S., & Froelicher, V. (1989). International application of a new probability algorithm for the diagnosis of coronary artery disease. *The American Journal of Cardiology*, 64(5), 304-310. [https://doi.org/10.1016/0002-9149\(89\)90524-9](https://doi.org/10.1016/0002-9149(89)90524-9)

4. **Little, M. A., McSharry, P. E., Hunter, E. J., & Ramig, L. O. (2009).** Suitability of dysphonia measurements for telemonitoring of Parkinson's disease. *IEEE Transactions on Biomedical Engineering*, 56(4), 1015-1022. <https://doi.org/10.1109/TBME.2008.2005954>
5. **Havlíček, V., Córcoles, A. D., Temme, K., Harrow, A. W., Kandala, A., Chow, J. M., & Gambetta, J. M. (2019).** Supervised learning with quantum-enhanced feature spaces. *Nature*, 567(7747), 209-212. <https://doi.org/10.1038/s41586-019-0980-2>
6. **Schuld, M., & Killoran, N. (2019).** Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122(4), 040504. <https://doi.org/10.1103/PhysRevLett.122.040504>
7. **Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., & Lloyd, S. (2017).** Quantum machine learning. *Nature*, 549(7671), 195-202. <https://doi.org/10.1038/nature23474>